

The KP Sehat Sahulat Program and Infant Mortality in Pakistan

A difference-in-differences evaluation of Pakistan's national health insurance program

Pakistan allocates just 3.38% of GDP to healthcare, and its infant mortality rate (56.9 deaths per 1,000 live births) ranks among the highest in the world. This paper evaluates the KP Sehat Sahulat Program — which insures poor households against health crises — in two stages: first, a regression identifying which household characteristics (household head's gender and education, household size, birth sequence) predict infant mortality; second, a difference-in-differences model using Khyber Pakhtunkhwa as the treatment group and Sindh/Punjab as controls to isolate the program's causal effect.

Amna Shahid · Econometrics Term Project, Group 1 · 2022

Led the econometric modeling, data analysis, and robustness testing for this team project (co-authored with Ubayd Qureshi, Masumeh Haider, and Maheen Ali Shah).

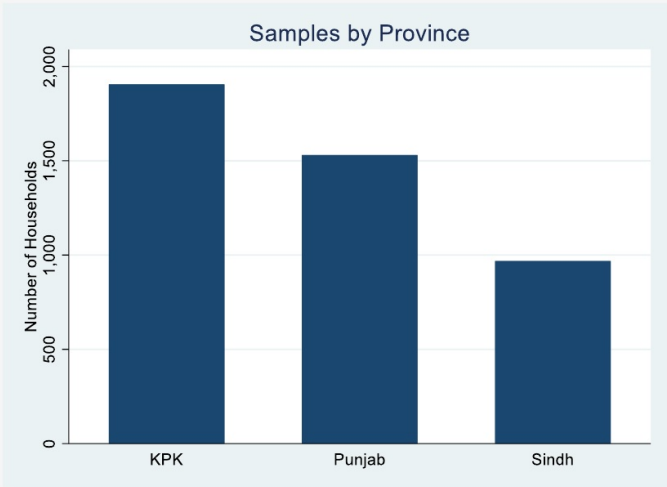
Understanding the program

The Sehat Sahulat Card was introduced in 2016 in Khyber Pakhtunkhwa (KPK) as part of the former Prime Minister's Sehat Sahulat Program — a social welfare reform giving poverty-stricken households access to medical care, including maternity services. Eligible families (Poverty Mean Score < 32.5) receive free treatment of up to Rs. 1 million in public and private hospitals.

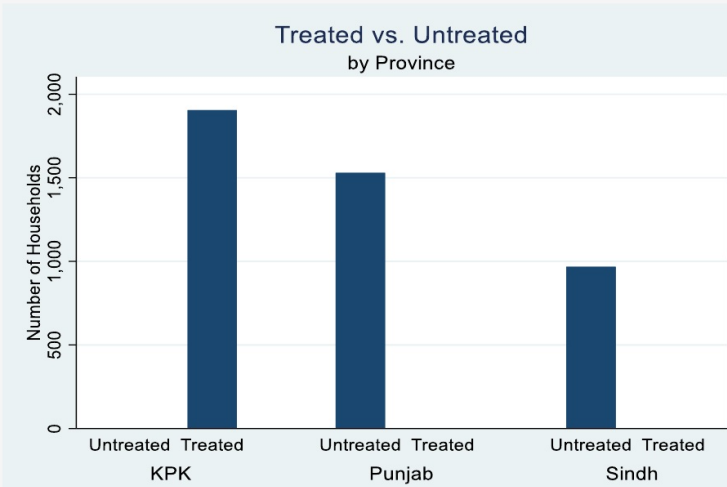
Data

The sample includes 4,403 household-level observations across three provinces. All KPK households in the sample had the Sahulat Card; none in Punjab or Sindh did, since the program had not yet been implemented there — making Punjab and Sindh a natural control group for isolating the program's effect in KPK.

Graph 1.Number of households by province

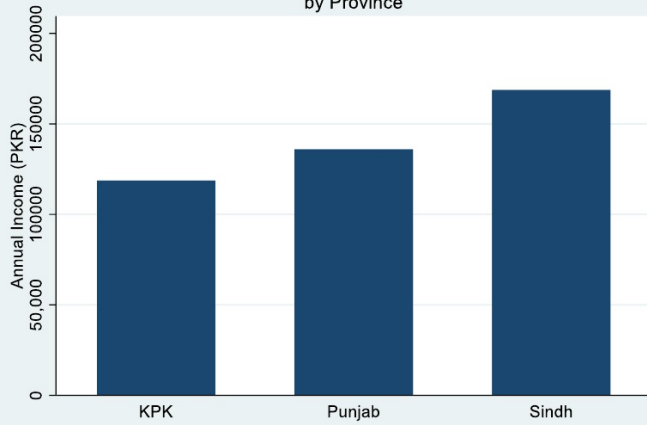


Graph 2.Treated vs Untreated Households



Sample composition by province, and treatment status by province.

Average Annual Household Income
by Province



Average annual household income is lowest in KPK and highest in Sindh — relevant since income is a plausible confound for both program eligibility and infant mortality.

Model 1: household predictors of infant mortality

A baseline regression tests how household head's gender and education, household size, and birth sequence relate to infant mortality — before isolating the program's specific effect.

Variable	Coefficient
Household head is male	0.0410*** (0.00914)
Household head's education (highest grade)	−0.00698** (0.00286)
Household size	−0.0113*** (0.00134)
Birth sequence	0.00794*** (0.00200)
Constant	0.0948*** (0.0126)
Observations / R²	4,403 / 0.018

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Male-headed households show **4.1 percentage points higher** infant mortality probability — consistent with prior literature suggesting mothers direct more resources toward child health and education. Each additional year of the household head's education is associated with a **0.7-point decrease**, and each step up in birth sequence with a **0.8-point increase**, reflecting strain on finite household resources. Counterintuitively, larger households show **lower** infant mortality (−1.1 points per additional member) — plausibly because more members means more earners or more caregivers.

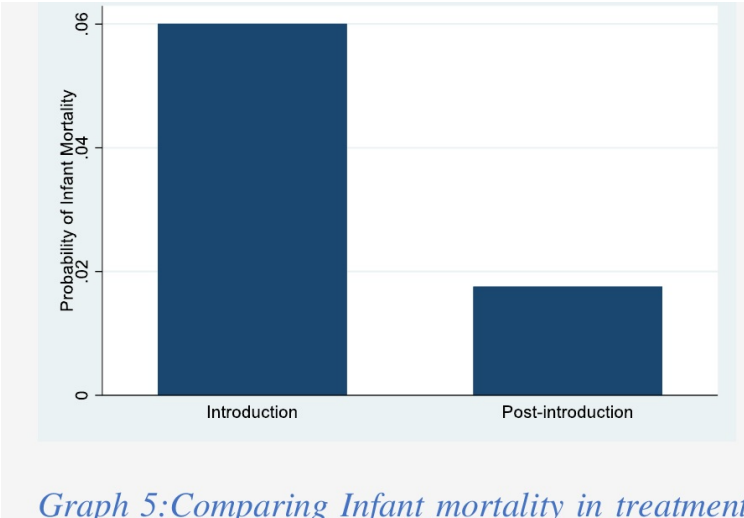
Model 2: difference-in-differences

The DiD design compares infant mortality in KP (treatment) against Sindh and Punjab (control), before and after the Sahulat card's introduction, isolating the program's causal effect from pre-existing trends.

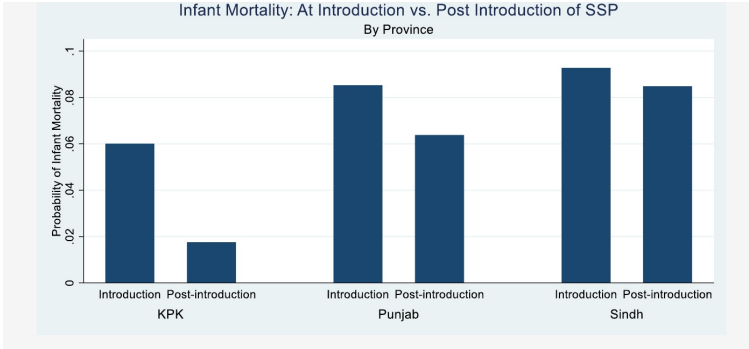
	Before (t=0)	After (t=1)	Difference
KP (treated)	$\beta_0 + \beta_1$	$\beta_0 + d_0 + \beta_1 + d_1$	$d_0 + d_1$
Sindh & Punjab (control)	β_0	$\beta_0 + d_0$	d_0
Difference-in-differences	β_1	$\beta_1 + d_1$	d_1

Variable	Coefficient
Time (post-treatment)	−0.0168 (0.0123)
Treat (KP)	−0.0282*** (0.00904)
DID (interaction term)	−0.0257* (0.0151)
Constant	0.0882*** (0.00654)
Observations / R ²	4,403 / 0.007

The DID coefficient of **−0.026**, significant at the 10% level, indicates the Sahulat card **reduced infant mortality in KP by roughly 2.6 percentage points** beyond what pre-existing trends would predict.



Graph 5:Comparing Infant mortality in treatment



Left: infant mortality in KP before vs. after the program. Right: the same comparison across all three provinces — KP's decline is the sharpest.

Robustness checks

Multicollinearity

Mean VIF across Model 1's predictors is **1.08** — close to 1, indicating multicollinearity is not a concern.

Omitted variable bias

A Ramsey RESET test rejects the null of no omitted variables ($F(3,4395) = 34.02$, $p < 0.001$) — expected, since the model omits variables like birth weight and public health infrastructure that weren't available in the dataset. Household income was considered but dropped in favor of education, which captures a similar underlying channel without duplicating it.

Heteroskedasticity

A Breusch-Pagan/Cook-Weisberg test rejects constant variance ($\chi^2(4) = 421.62$, $p < 0.001$) — expected given infant mortality is a binary outcome modeled as a linear probability. Robust standard errors were used throughout to correct for this; coefficients remained similarly significant.

Limitations

The program was not randomly assigned — it targeted households below the poverty line, introducing selection bias. Provincial spending differences may mean KP and the control provinces didn't share a true parallel pre-trend even before treatment, so not all of the estimated difference can be attributed to the program alone. Behavioral factors (e.g. preference for home births) and practical frictions (travel cost to reach card distribution centers) may also limit how fully eligible households actually use the benefit.

Conclusion

The KP Sehat Sahulat Program appears to have meaningfully reduced infant mortality in the province. Household-level factors — head's gender and education, household size, and birth sequence — all significantly predict infant mortality, suggesting targeted policy levers beyond insurance coverage alone. Programs like SSP can meaningfully advance Pakistan's progress toward WHO's Universal Health Coverage commitments.

Selected references

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